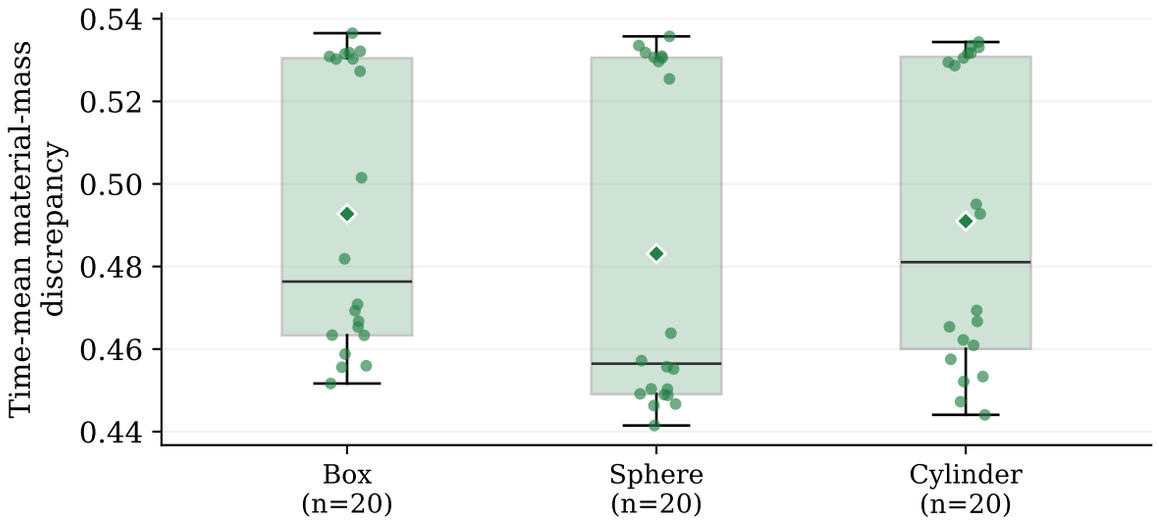

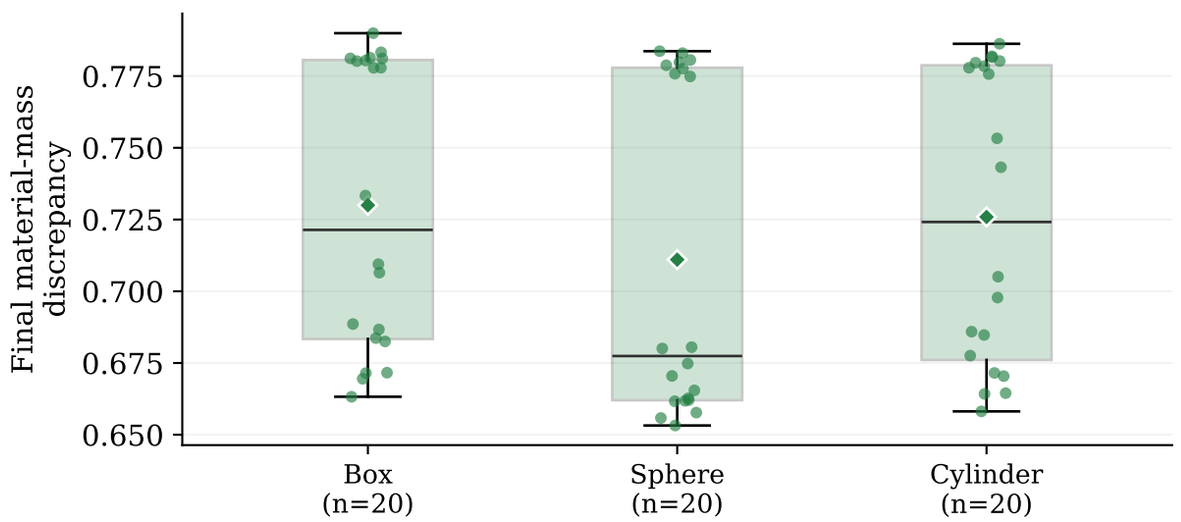

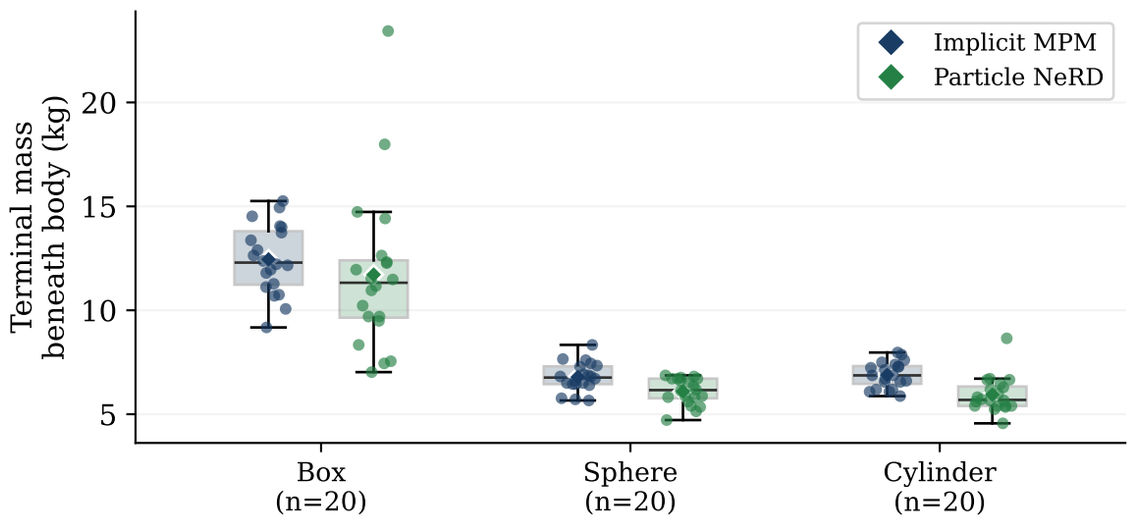

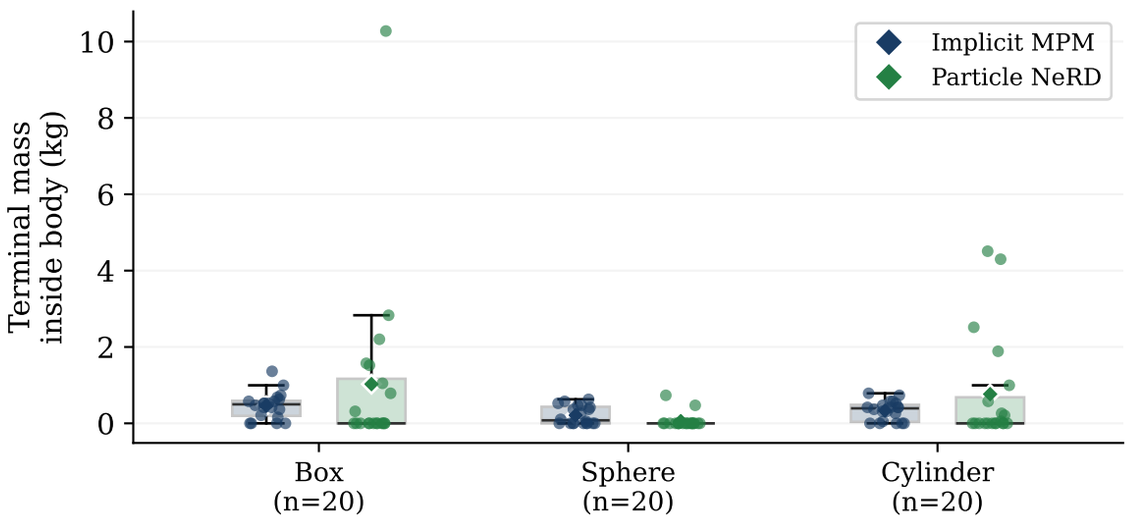

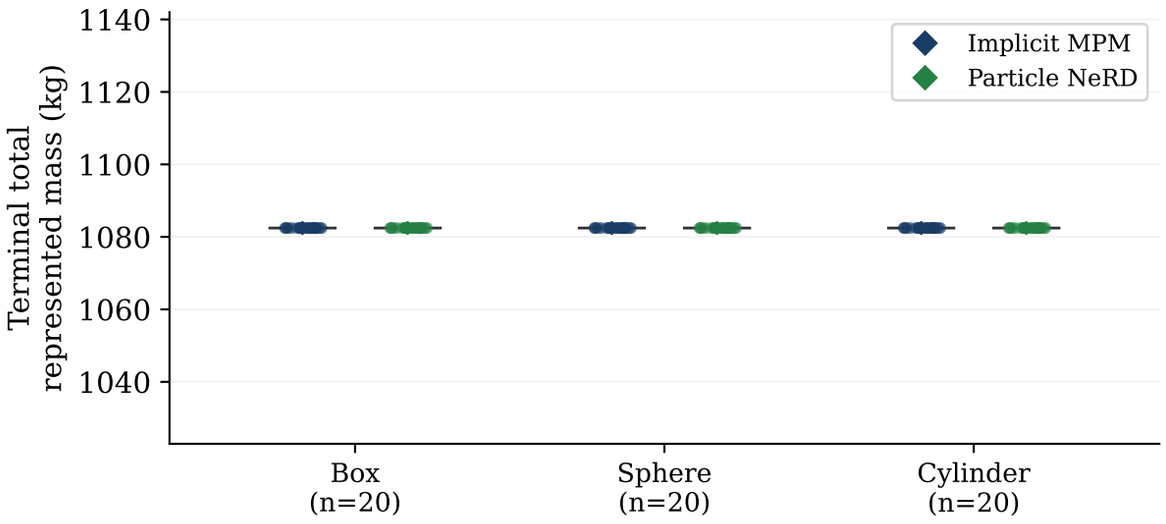

Figure 3: Frozen object-model checkpoint 9,000. The first row shows time-mean and terminal-state material-mass discrepancy: sum of absolute cell-mass differences divided by total physical mass. The second row shows terminal-state mass beneath and inside the body; the last panel shows total represented mass. Terminal state is 6 s. Equal total mass does not imply correct distribution or confinement to the tray. Each dot is one trajectory summary; boxes show the median and interquartile range, whiskers extend to 1.5 IQR, diamonds mark means. Twenty trajectories per shape unless labeled otherwise. Time means use available samples; empty particle regions are not assigned zero error.